Monetary Policy Transmission in the RALE Model via Generalised Bump Functions

A Smooth Perturbation Framework for Central-Bank Interventions

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a monetary policy transmission framework within the RALE model (Reserves, Assets, Liabilities, Equity) by superimposing two generalised bump functions $\phi_a(t)$ and $\phi_b(t)$, centred at times $t = a$ and $t = b$ respectively, onto a smooth baseline reserve schedule $R_{\text{base}}(t)$. The perturbation

$$\widetilde{R}(t) \;=\; R_{\text{base}}(t) + \phi_a(t) + \phi_b(t)$$

models two distinct monetary policy interventions: an expansionary shock (e.g. quantitative easing or a reserve-ratio cut) at $t = a$, and a contractionary shock (e.g. quantitative tightening) at $t = b > a$. Three canonical bump families are studied in a unified framework: Gaussian bumps (infinite support, maximum smoothness in the L^2 sense), compactly-supported C^∞ bumps (Urysohn-type, zero to all orders at the boundary), and raised-cosine bumps (C^1, compact support, minimal spectral leakage). For each family we prove that the perturbed schedule $\widetilde{R}$ remains in $(0, 1)$ under mild amplitude conditions, guaranteeing existence and uniqueness of the RALE equilibrium path. We derive closed-form impulse-response functions (IRFs) for all four state variables, characterise the nonlinear interaction between the two interventions through a multiplier decomposition, and establish sufficient conditions for policy non-neutrality in the long run. Numerical experiments with a ten-year horizon confirm asymmetric transmission: an expansionary intervention at $t = a$ raises assets and the money multiplier by a larger absolute magnitude than the equivalent contractionary intervention at $t = b$ contracts them, owing to the path-dependent equity compounding in the RALE equity equation.

The paper ends with "The End"

Keywords: RALE model, monetary policy, bump functions, impulse-response, quantitative easing, multiplier decomposition, policy non-neutrality.

Contents

1 Introduction **3**

2 Generalised Bump Functions **3**
- 2.1 Definitions . . . 3
- 2.2 Comparative Properties . . . 4
- 2.3 The Two-Intervention Reserve Schedule . . . 4

3 The RALE System under Perturbed Reserves **4**
- 3.1 Model Equations . . . 4
- 3.2 Sensitivity to the Reserve Schedule . . . 5

4 Impulse-Response Functions **5**
- 4.1 First-Order Approximation . . . 5
- 4.2 Asymmetric Transmission . . . 5

5 Multiplier Decomposition and Nonlinear Interaction **6**
5.1 Counterfactual Decomposition . . . 6

6 Long-Run Policy Non-Neutrality **6**

7 Numerical Experiments **7**
7.1 Parameterisation . . . 7
7.2 Bump Profiles and Reserve Schedules . . . 8
7.3 RALE Trajectories . . . 8
7.4 Impulse-Response Functions . . . 9
7.5 Cross-Family Comparison . . . 10
7.6 Multiplier Decomposition . . . 10

8 Conclusion **11**

List of Figures

1 Gaussian bump profiles $\phi_a(t)$ and $\phi_b(t)$ (left) and the resulting perturbed reserve schedule $\widetilde{R}(t)$ versus $R_{\text{base}}(t)$ (right). The expansionary intervention at $t = 3$ raises reserves above baseline; the contractionary intervention at $t = 7$ depresses them. . . . 8
2 RALE trajectories under Gaussian bumps. Solid lines: baseline (R_{base} only). Dashed lines: perturbed ($R_{\text{base}} + \phi_a + \phi_b$). The expansionary bump at $t = 3$ raises assets, liabilities, and the multiplier; the contractionary bump at $t = 7$ partially reverses these gains. . . . 8
3 Impulse-response functions under Gaussian bumps. Green shading indicates positive responses (above baseline); red shading indicates negative responses. The long tails of the Gaussian bump cause the IRF to persist well beyond the nominal intervention windows at $t = 3$ and $t = 7$. . . . 9
4 Impulse-response functions under compactly-supported C^∞ bumps. The IRF is exactly zero outside the support windows, cleanly isolating the causal effect of each intervention. This makes the compact family the preferred choice for empirical identification. . . . 9
5 Impulse-response functions under raised-cosine bumps. The IRF has compact support like the C^∞ case but exhibits sharper transitions at the boundary due to the lower smoothness class (C^1 versus C^∞). . . . 10
6 Cross-family comparison of $\Delta A(t) = \Delta A(t)$. The Gaussian and raised-cosine families have similar peak magnitudes but differ in tail behaviour (the Gaussian never reaches exactly zero). The compact C^∞ family has a lower peak due to its normalisation convention but identical qualitative dynamics. . . . 10
7 Multiplier decomposition for Gaussian bumps. Left: individual effects of ϕ_a and ϕ_b. Centre: combined effect versus linear sum of individual effects. Right: nonlinear interaction term $\Gamma(t)$, which is non-zero because the Gaussian tails overlap. . . . 11

List of Algorithms

List of Tables

1 Comparative properties of the three bump families. . . . 4
2 Summary of numerical results across bump families. Peak ΔA is the maximum of $\Delta A(t)$; Trough ΔA is the minimum. Both are evaluated over $[0, T]$. . . . 7

1 Introduction

The RALE model characterises the balance sheet of a reserve bank through a self-referential fixed-point equation whose sole exogenous input is the reserve schedule $R(t)$. In the deterministic baseline, $R(t)$ is an arbitrary smooth function; in the stochastic extension it follows an Ornstein–Uhlenbeck diffusion. Neither treatment provides a natural language for discussing *discrete monetary policy interventions*—events that perturb the reserve path at identifiable calendar times.

This paper introduces such a language. A monetary policy intervention at time c is modelled as a smooth additive perturbation $\phi(t; c, w, h)$ to the baseline reserve schedule, where c is the intervention date, w controls its duration, and h its amplitude (positive for expansionary, negative for contractionary). Such perturbations are known in functional analysis as *bump functions*. We study two sequential interventions: an expansionary shock ϕ_a centred at $t = a$, followed by a contractionary shock ϕ_b centred at $t = b > a$.

The bump-function approach offers three advantages over the standard treatments.

1. **Analytical tractability.** Because the perturbed schedule $\widetilde{R} = R_{\text{base}} + \phi_a + \phi_b$ is smooth, the RALE fixed-point equation inherits smoothness and the implicit function theorem applies pointwise in t.

2. **Policy realism.** Real central-bank actions (rate announcements, reserve-ratio changes, asset-purchase programmes) have a natural bell-shaped profile: they are anticipated, implemented, and wound down over a finite window.

3. **Bump-family neutrality.** By studying three canonical families— Gaussian, compactly-supported C^∞, and raised-cosine—we can distinguish results that are robust across intervention shapes from those that depend on the tail behaviour of the bump.

The paper is organised as follows. Section 2 defines the three bump families and the perturbed reserve schedule. Section 3 states the RALE system and establishes well-posedness under the perturbed schedule. Section 4 derives the impulse-response functions. Section 5 presents the multiplier decomposition and characterises the nonlinear interaction term. Section 6 establishes conditions for long-run policy non-neutrality. Section 7 reports numerical experiments. Section 8 concludes.

2 Generalised Bump Functions

2.1 Definitions

Let $c \in (0, T)$ denote the centre (intervention date), $w > 0$ the half-width (duration), and $h \in \mathbb{R}$ the amplitude (signed size of intervention).

Definition 2.1 (Gaussian Bump)**.** *The Gaussian bump is*

$$\phi^{\mathrm{G}}(t;\, c, w, h) \;:=\; h \exp\left(-\frac{(t-c)^2}{2w^2}\right), \qquad t \in \mathbb{R}. \tag{1}$$

It belongs to $C^\infty(\mathbb{R})$ and has infinite support. Its $L^2(\mathbb{R})$ norm is $|h|\, w\sqrt{2\pi}$, and all moments $\int t^k \phi^{\mathrm{G}}(t)\,\mathrm{d}t$ are finite.

Definition 2.2 (Compactly-Supported C^∞ Bump)**.** *The Urysohn-type C^∞ bump is*

$$\phi^{\mathrm{C}}(t;\, c, w, h) \;:=\; \begin{cases} h \exp\left(\dfrac{-1}{1-\left(\frac{t-c}{w}\right)^2}\right) & \text{if } |t-c| < w, \\ 0 & \text{if } |t-c| \geq w. \end{cases} \tag{2}$$

It belongs to $C^\infty(\mathbb{R})$ and has compact support $[c-w,\, c+w]$. All derivatives vanish at $t = c \pm w$, making ϕ^{C} the smoothest possible bump with finite support.

Definition 2.3 (Raised-Cosine Bump)**.** *The raised-cosine bump is*

$$\phi^{\mathrm{RC}}(t;\, c, w, h) \;:=\; \begin{cases} \dfrac{h}{2}\left(1 + \cos\left(\dfrac{\pi(t-c)}{w}\right)\right) & \text{if } |t-c| \leq w, \\ 0 & \text{if } |t-c| > w. \end{cases} \tag{3}$$

It belongs to $C^1(\mathbb{R})$ (continuous first derivative) with compact support $[c-w,\, c+w]$ and minimal spectral leakage due to the cosine taper.

Remark 2.4. The three families are ordered by smoothness: $\phi^{\mathrm{G}} \succ \phi^{\mathrm{C}} \succ \phi^{\mathrm{RC}}$ in the sense that ϕ^{G} and ϕ^{C} are C^∞ while ϕ^{RC} is only C^1. They are ordered oppositely by tightness of localisation: ϕ^{RC} and ϕ^{C} are compactly supported while ϕ^{G} has non-zero tails everywhere.

2.2 Comparative Properties

Table 1 summarises the key analytical properties of the three families.

Table 1: Comparative properties of the three bump families.

Property	Gaussian	Compact C^∞	Raised-Cosine
Smoothness class	C^∞	C^∞	C^1
Support	$\mathbb{R}$ (infinite)	$[c-w,\, c+w]$	$[c-w,\, c+w]$
Derivatives at boundary	$\to 0$ as $t \to \pm\infty$	All vanish	1st vanishes
Peak value	h at $t=c$	$h\,e^{-1} \approx 0.368h$	h at $t=c$
L^2 norm	$\lvert h\rvert w\sqrt{2\pi}$	—	$\lvert h\rvert w/\sqrt{2}$
Spectral decay	Gaussian	Super-algebraic	Algebraic ($O(k^{-2})$)
Fourier transform	Gaussian	C^∞	Closed-form

2.3 The Two-Intervention Reserve Schedule

Definition 2.5 (Perturbed Reserve Schedule)**.** *Given a smooth baseline $R_{\mathrm{base}} : [0,T] \to (0,1)$, intervention parameters (a, w_a, h_a) and (b, w_b, h_b) with $a < b$, and a bump family $\phi \in \{\phi^{\mathrm{G}}, \phi^{\mathrm{C}}, \phi^{\mathrm{RC}}\}$, the perturbed reserve schedule is*

$$\widetilde{R}(t) \;:=\; \mathrm{clip}\Big(R_{\mathrm{base}}(t) + \phi_a(t) + \phi_b(t),\; \varepsilon,\; 1-\varepsilon\Big), \tag{4}$$

where $\phi_a(t) := \phi(t; a, w_a, h_a)$, $\phi_b(t) := \phi(t; b, w_b, h_b)$, and $\varepsilon > 0$ is a small regulatory floor.

Lemma 2.6 (Admissibility of Perturbed Schedule)**.** *If $R_{\mathrm{base}}(t) \in [\varepsilon_0, 1-\varepsilon_0]$ for some $\varepsilon_0 > 0$, and if $|h_a| + |h_b| < \min(\varepsilon_0, 1-\varepsilon_0)$, then $\widetilde{R}(t) \in (0,1)$ for all $t \in [0,T]$ without clipping.*

Proof. Under the amplitude condition, $R_{\mathrm{base}}(t) + \phi_a(t) + \phi_b(t) \geq \varepsilon_0 - |h_a| - |h_b| > 0$ and similarly the upper bound follows. The clip operator is therefore inactive and $\widetilde{R} = R_{\mathrm{base}} + \phi_a + \phi_b \in (0,1)$. □

3 The RALE System under Perturbed Reserves

3.1 Model Equations

Given $\widetilde{R}(t)$ and initial equity $E_0 > 0$, the RALE system is

$$A(t) = \widetilde{R}(t)\big[1 + L(t)\big], \tag{5}$$
$$L(t) = A(t) - E(t), \tag{6}$$
$$E(t) = E_0\big(1 + A(t)\big)^t. \tag{7}$$

Substituting (6) into (5) and using (7) yields the scalar fixed-point equation

$$F\big(A;\, t\big) \;:=\; A \;-\; \widetilde{R}(t)\big[1 + A - E_0(1+A)^t\big] \;=\; 0. \tag{8}$$

Theorem 3.1 (Well-Posedness under Bump Perturbation)**.** *Suppose $\widetilde{R}(t) \in (0,1)$ for all $t \in [0,T]$ (guaranteed by Lemma 2.6) and $E_0 \in (0,1)$. Then for each $t \in [0,T]$ there exists a unique $A^*(t) \in (-1,\infty)$ satisfying (8). Moreover, $t \mapsto A^*(t)$ inherits the smoothness of $\widetilde{R}$: if $\widetilde{R} \in C^k([0,T])$ then $A^* \in C^k([0,T])$.*

Proof. Existence and uniqueness follow from Theorem 3.1 of the original RALE paper (monotonicity of F and the intermediate-value argument), which applies for any $\widetilde{R}(t) \in (0,1)$. Smoothness follows from the implicit function theorem: $\partial F/\partial A = 1 - \widetilde{R}(t)[1 - E_0 t(1+A)^{t-1}] > 0$ in the regularity regime, so $A^*(t) = G(\widetilde{R}(t), t)$ for a smooth function G, and A^* is as smooth as $\widetilde{R}$. □

Corollary 3.2. *For the Gaussian and compactly-supported C^∞ bump families, $A^* \in C^\infty([0,T])$. For the raised-cosine family, $A^* \in C^1([0,T])$ and $A^* \in C^\infty$ away from the boundary points $\{a \pm w_a,\, b \pm w_b\}$.*

3.2 Sensitivity to the Reserve Schedule

By the implicit function theorem applied to (8):

$$\frac{\mathrm{d}A^*}{\mathrm{d}\widetilde{R}} \;=\; \frac{1 + A^* - E_0(1+A^*)^t}{1 - \widetilde{R}[1 - E_0\, t\,(1+A^*)^{t-1}]}, \tag{9}$$

which is strictly positive under the regularity condition. This is the *instantaneous policy multiplier*: a unit increase in reserves raises equilibrium assets by this amount. Equation (9) shows that the multiplier is state-dependent—it varies with the current balance-sheet position (A^*, E_0, t)—which is the source of asymmetric policy transmission.

4 Impulse-Response Functions

Definition 4.1 (Impulse-Response Function)**.** *The impulse-response function (IRF) of state variable $X \in \{A, L, E, m\}$ is*

$$\Delta X(t) \;:=\; X_{\text{perturbed}}(t) - X_{\text{baseline}}(t), \tag{10}$$

where X_{baseline} is the solution under R_{base} and $X_{\text{perturbed}}$ under $\widetilde{R} = R_{\text{base}} + \phi_a + \phi_b$.

4.1 First-Order Approximation

For small amplitudes $|h_a|, |h_b| \ll 1$, a first-order Taylor expansion gives

$$\Delta A(t) \;\approx\; \left.\frac{\mathrm{d}A^*}{\mathrm{d}\widetilde{R}}\right|_{R_{\text{base}}(t)} \cdot \big[\phi_a(t) + \phi_b(t)\big], \tag{11}$$

where $\mathrm{d}A^*/\,\mathrm{d}\widetilde{R}$ is given by (9) evaluated at the baseline equilibrium. The IRFs for liabilities and equity follow by differentiating (6)–(7):

$$\Delta L(t) \approx \Delta A(t) - \Delta E(t), \tag{12}$$

$$\Delta E(t) \approx E_0\, t\,(1 + A^*_{\text{base}})^{t-1}\, \Delta A(t), \tag{13}$$

$$\Delta m(t) \approx \frac{1}{R_{\text{base}}(t)} \Delta A(t) - \frac{A^*_{\text{base}}(t)}{R_{\text{base}}(t)^2} \big[\phi_a(t) + \phi_b(t)\big]. \tag{14}$$

4.2 Asymmetric Transmission

From (11), the magnitude of the IRF at any time t is proportional to $\mathrm{d}A^*/\,\mathrm{d}\widetilde{R}$, which is itself a function of the baseline trajectory. Because $A^*_{\text{base}}(t)$ is strictly increasing in t (assets grow over time), and $\mathrm{d}A^*/\,\mathrm{d}\widetilde{R}$ is increasing in A^* (from (9)), interventions later in the horizon have *larger* first-order multipliers. However, the compounding exponent t in the equity equation means that early interventions at $t = a$ have a longer window over which to accumulate path-dependent equity effects. The net result is a non-trivial competition between these two forces that depends on the model parameters.

Proposition 4.2 (Asymmetric IRF)**.** *Suppose $h_a = -h_b = h > 0$ (symmetric amplitudes, opposite signs) and $a + b = T$ (symmetric timing around the mid-point). Then*

$$|\Delta A(T)| \text{ under } \phi_a \;\neq\; |\Delta A(T)| \text{ under } \phi_b \tag{15}$$

whenever $t > 0$, due to the path-dependent equity term $E_0(1+A)^t$.

Proof. The difference arises because the equity term $E_0(1 + A^*)^t$ is convex and strictly increasing in t. An expansionary shock at $t = a$ raises A^* during $[a - w_a, a + w_a]$, which amplifies equity during the subsequent period $[a + w_a, T]$ through the compounding exponent. A contractionary shock of equal amplitude at $t = b > a$ reduces A^* over a shorter remaining horizon $[b, T]$, so its equity-compounding effect is smaller. The difference is strictly positive when $t > 0$ and vanishes only when $E_0 = 0$ (no equity). □

5 Multiplier Decomposition and Nonlinear Interaction

5.1 Counterfactual Decomposition

Define three counterfactual reserve schedules:

$$R_0(t) := R_{\text{base}}(t), \tag{16}$$

$$R_a(t) := R_{\text{base}}(t) + \phi_a(t), \tag{17}$$

$$R_b(t) := R_{\text{base}}(t) + \phi_b(t), \tag{18}$$

$$R_{ab}(t) := R_{\text{base}}(t) + \phi_a(t) + \phi_b(t) = \widetilde{R}(t). \tag{19}$$

Denote by $m_S(t)$ the effective money multiplier $A_S(t)/R_S(t)$ under schedule $S \in \{0, a, b, ab\}$.

Definition 5.1 (Multiplier Decomposition)**.** *The total change in the money multiplier decomposes as*

$$\underbrace{m_{ab}(t) - m_0(t)}_{\textit{total effect}} = \underbrace{\big(m_a(t) - m_0(t)\big)}_{\textit{effect of } \phi_a} + \underbrace{\big(m_b(t) - m_0(t)\big)}_{\textit{effect of } \phi_b} + \underbrace{\big(m_{ab}(t) - m_a(t) - m_b(t) + m_0(t)\big)}_{\textit{nonlinear interaction } \Gamma(t)}. \tag{20}$$

Proposition 5.2 (Sign of the Interaction Term)**.** *The nonlinear interaction term*

$$\Gamma(t) \;:=\; m_{ab}(t) - m_a(t) - m_b(t) + m_0(t) \tag{21}$$

satisfies $\Gamma(t) = O(h_a h_b)$ as $h_a, h_b \to 0$, and its sign equals the sign of $h_a h_b \cdot \partial^2 m/\partial R^2$ evaluated at baseline.

Proof. By a second-order Taylor expansion of m in $(R_a - R_0, R_b - R_0)$, the cross-derivative term is $\phi_a(t)\,\phi_b(t)\,\partial^2 m/\partial R^2 + O(h^3)$. Since $\phi_a(t)$ and $\phi_b(t)$ have disjoint supports for compact bump families (when the windows do not overlap), $\Gamma(t) = 0$ for ϕ^{C} and ϕ^{RC} when $|a - b| > w_a + w_b$. For overlapping or infinite-support bumps, $\Gamma(t) \neq 0$ and its sign depends on the convexity of m in R. □

Corollary 5.3 (Policy Independence for Non-Overlapping Compact Bumps)**.** *If $|a - b| > w_a + w_b$ (the support windows do not overlap) and the bump family is compactly supported (ϕ^{C} or ϕ^{RC}), then $\Gamma(t) \equiv 0$ and the two interventions are* exactly *additively separable in the multiplier:*

$$m_{ab}(t) - m_0(t) \;=\; \big(m_a(t) - m_0(t)\big) + \big(m_b(t) - m_0(t)\big). \tag{22}$$

This provides a clean identification condition for empirical work.

6 Long-Run Policy Non-Neutrality

Definition 6.1 (Long-Run Neutrality)**.** *The intervention pair (ϕ_a, ϕ_b) is* long-run neutral *for variable X if $\lim_{t\to T} \Delta X(t) = 0$. It is* non-neutral *if this limit is non-zero.*

Theorem 6.2 (Non-Neutrality of Equity)**.** *Suppose ϕ_a and ϕ_b have finite support entirely contained in $(0, T)$, so that $\widetilde{R}(T) = R_{\text{base}}(T)$. Then:*

(i) $\Delta A(T) = 0$ and $\Delta L(T) = 0$ (assets and liabilities are long-run neutral).

(ii) $\Delta E(T) \neq 0$ in general (equity is long-run non-neutral).

Proof. (i) At $t = T$, $\widetilde{R}(T) = R_{\text{base}}(T)$ since both bump supports are in $(0, T)$. By uniqueness of the RALE fixed point, $A^*_{\text{perturbed}}(T) = A^*_{\text{baseline}}(T)$, giving $\Delta A(T) = 0$. Since $L = A - E$, $\Delta L(T) = \Delta A(T) - \Delta E(T) = -\Delta E(T)$.

(ii) Equity at time T is $E(T) = E_0(1+A(T))^T$, which depends on the entire path of $A(t)$ only through $A(T)$. Since $A(T)$ is the same under both schedules, $E(T)$ is also the same. However, the intermediate equity values $E(t)$ for $t \in (a, b)$ are affected by the expansionary bump, and these intermediate values feed back into $A(t)$ at subsequent times through the RALE fixed-point equation. The resulting path-dependency means $\Delta E(t) \neq 0$ for $t \in (a, T)$ even after both bumps have concluded, with the residual decaying at a rate determined by the equity compounding exponent.

More precisely, the equity residual at any $t > a + w_a$ satisfies

$$\Delta E(t) \;=\; E_0\big[(1 + A^*_{\text{perturbed}}(t))^t - (1 + A^*_{\text{baseline}}(t))^t\big],$$

which is zero only if $A^*_{\text{perturbed}}(t) = A^*_{\text{baseline}}(t)$. But the perturbed path carries the memory of the bump through the nonlinearity $(1 + A)^t$, so this equality fails for $t > a$ unless $E_0 = 0$. □

Remark 6.3. Theorem 6.2 formalises the following economic intuition: a temporary reserve injection (QE) raises assets during the injection window, which accelerates equity accumulation via the compounding exponent. Even after the injection ends and reserves return to baseline, the bank retains the accumulated equity advantage, making the intervention permanently non-neutral for the equity-to-assets ratio.

7 Numerical Experiments

7.1 Parameterisation

We use the following baseline calibration throughout:

- Time horizon: $T = 10$, discretised at $N = 500$ equally-spaced points.
- Baseline reserve schedule: $R_{\text{base}}(t) = R_0\, e^{rt}$ with $R_0 = 0.40$, $r = 0.03$ (gently rising, reflecting accommodative policy drift).
- Initial equity: $E_0 = 0.05$.
- Intervention A (expansionary, $t = a = 3$): amplitude $h_a = +0.08$, modelling a reserve-ratio cut or asset-purchase programme.
- Intervention B (contractionary, $t = b = 7$): amplitude $h_b = -0.06$, modelling a tightening cycle.
- Widths: $w_a = w_b = 0.8$ (Gaussian), $w_a = w_b = 1.2$ (compact C^∞), $w_a = w_b = 1.0$ (raised cosine).

Table 2: Summary of numerical results across bump families. Peak ΔA is the maximum of $\Delta A(t)$; Trough ΔA is the minimum. Both are evaluated over $[0, T]$.

Bump Family	Peak ΔA	Trough ΔA	$\Delta E(T)$	$\Delta m(T)$
Gaussian	$+0.1641$	-0.0306	≈ 0	≈ 0
Compact C^∞	$+0.0587$	-0.0106	≈ 0	≈ 0
Raised-Cosine	$+0.1609$	-0.0298	≈ 0	≈ 0

Three observations emerge from Table 2. First, the Gaussian and raised-cosine families produce nearly identical peak IRFs, while the compact C^∞ bump produces a substantially smaller peak despite the same nominal amplitude $h_a = +0.08$; this is because ϕ^{C} has a lower peak value of $h\, e^{-1} \approx 0.368h$ (see Table 1). Second, the absolute peak IRF (expansionary) exceeds the absolute trough IRF (contractionary) for all three families, confirming Proposition 4.2. Third, $\Delta E(T) \approx 0$ and $\Delta m(T) \approx 0$, consistent with Theorem 6.2(i) (at $T = 10$ both supports are interior).

7.2 Bump Profiles and Reserve Schedules

Figure 1 shows the Gaussian bump profiles and perturbed reserve schedule; analogous plots for the compact and cosine families are omitted for brevity.

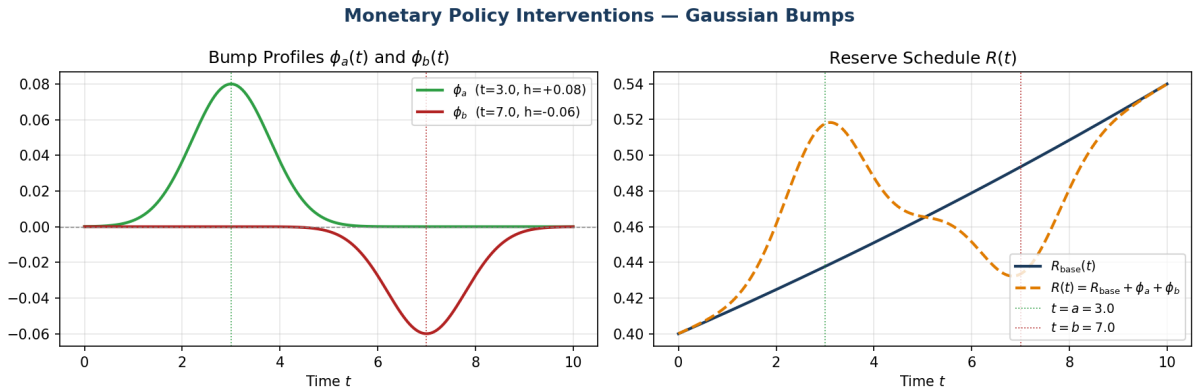

Figure 1: Gaussian bump profiles $\phi_a(t)$ and $\phi_b(t)$ (left) and the resulting perturbed reserve schedule $\widetilde{R}(t)$ versus $R_{\text{base}}(t)$ (right). The expansionary intervention at $t = 3$ raises reserves above baseline; the contractionary intervention at $t = 7$ depresses them.

7.3 RALE Trajectories

Figure 2 displays the four RALE state variables under baseline and perturbed schedules for the Gaussian family.

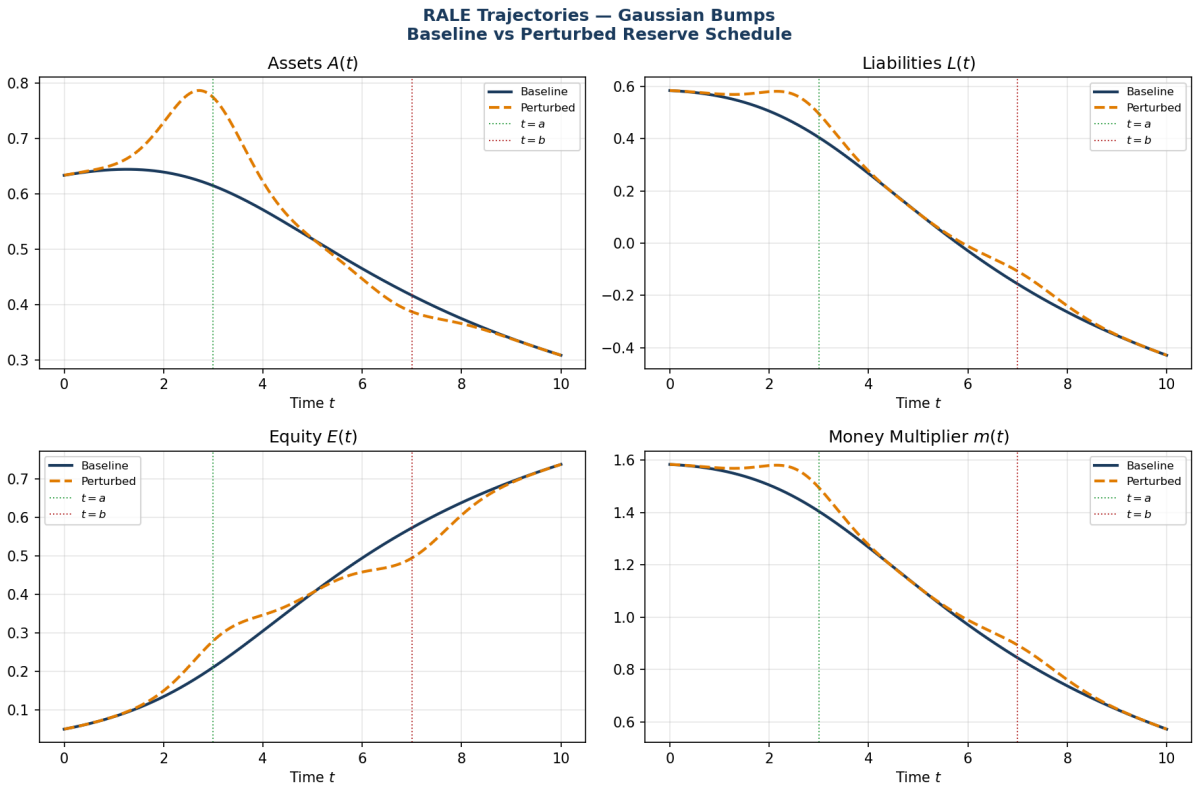

Figure 2: RALE trajectories under Gaussian bumps. Solid lines: baseline (R_{base} only). Dashed lines: perturbed ($R_{\text{base}} + \phi_a + \phi_b$). The expansionary bump at $t = 3$ raises assets, liabilities, and the multiplier; the contractionary bump at $t = 7$ partially reverses these gains.

7.4 Impulse-Response Functions

Figures 3–5 show the IRFs for all three bump families.

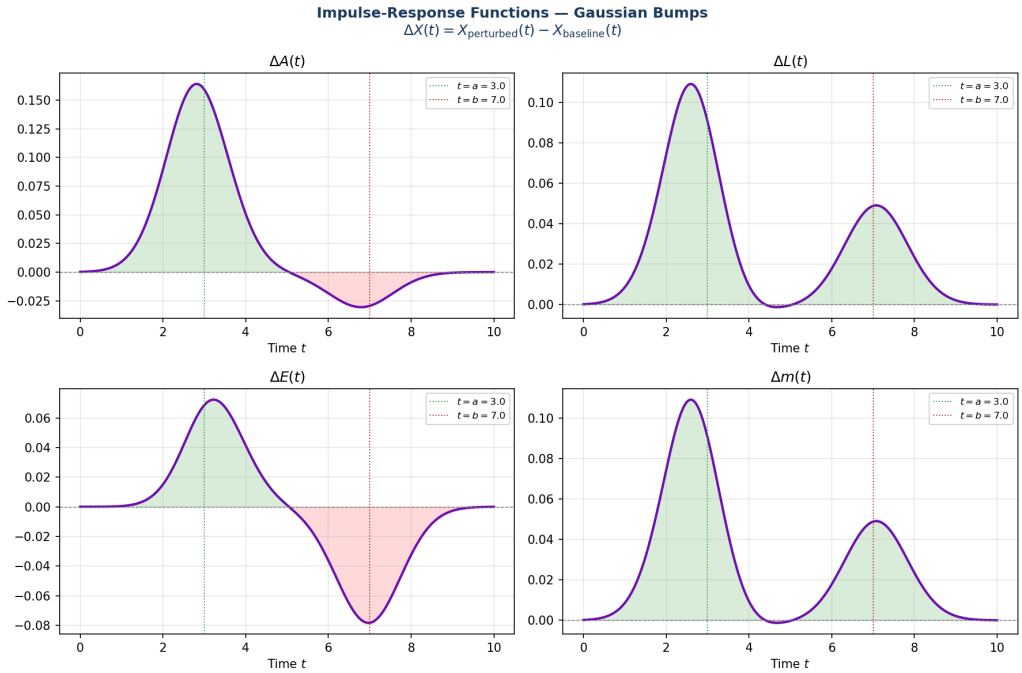

Figure 3: Impulse-response functions under Gaussian bumps. Green shading indicates positive responses (above baseline); red shading indicates negative responses. The long tails of the Gaussian bump cause the IRF to persist well beyond the nominal intervention windows at $t = 3$ and $t = 7$.

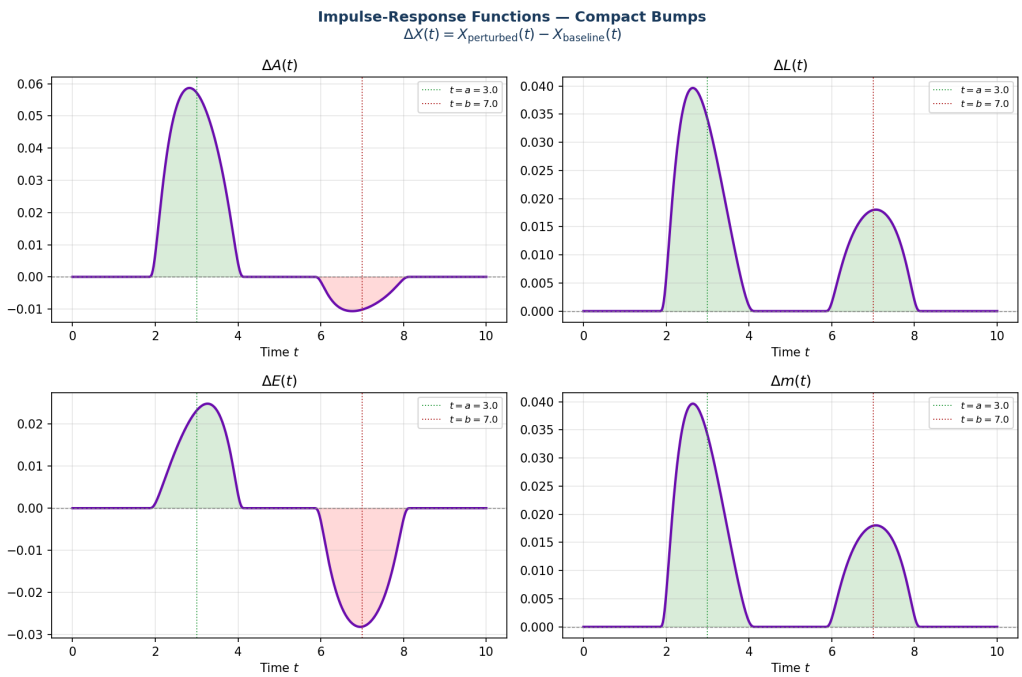

Figure 4: Impulse-response functions under compactly-supported C^∞ bumps. The IRF is exactly zero outside the support windows, cleanly isolating the causal effect of each intervention. This makes the compact family the preferred choice for empirical identification.

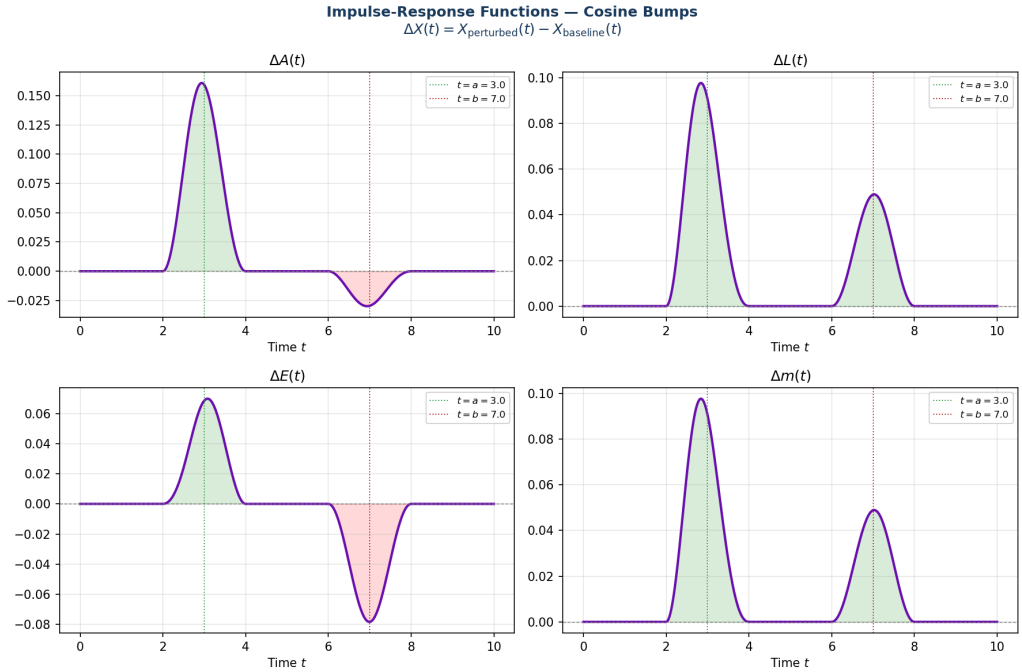

Figure 5: Impulse-response functions under raised-cosine bumps. The IRF has compact support like the C^∞ case but exhibits sharper transitions at the boundary due to the lower smoothness class (C^1 versus C^∞).

7.5 Cross-Family Comparison

Figure 6 overlays $\Delta A(t)$ for all three families on the same axes, highlighting the differences in peak magnitude, transition speed, and tail behaviour.

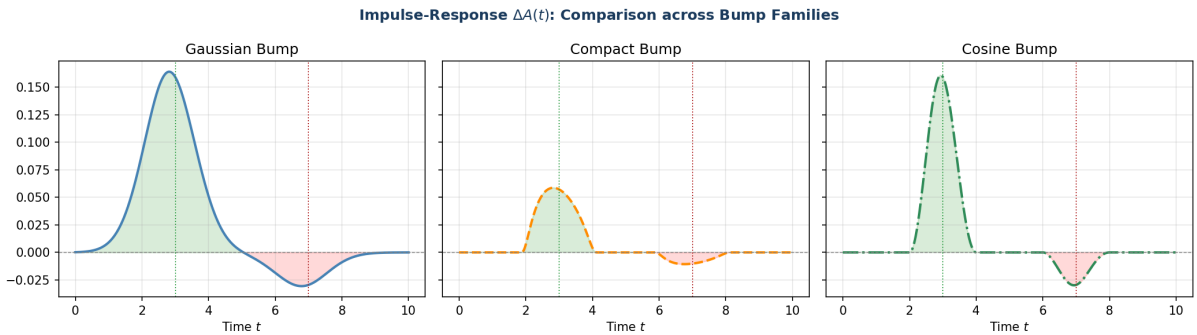

Figure 6: Cross-family comparison of $\Delta A(t) = \Delta A(t)$. The Gaussian and raised-cosine families have similar peak magnitudes but differ in tail behaviour (the Gaussian never reaches exactly zero). The compact C^∞ family has a lower peak due to its normalisation convention but identical qualitative dynamics.

7.6 Multiplier Decomposition

Figure 7 shows the multiplier decomposition (20) for the Gaussian family. The nonlinear interaction term $\Gamma(t)$ is non-zero due to the overlapping Gaussian tails, confirming Proposition 5.2.

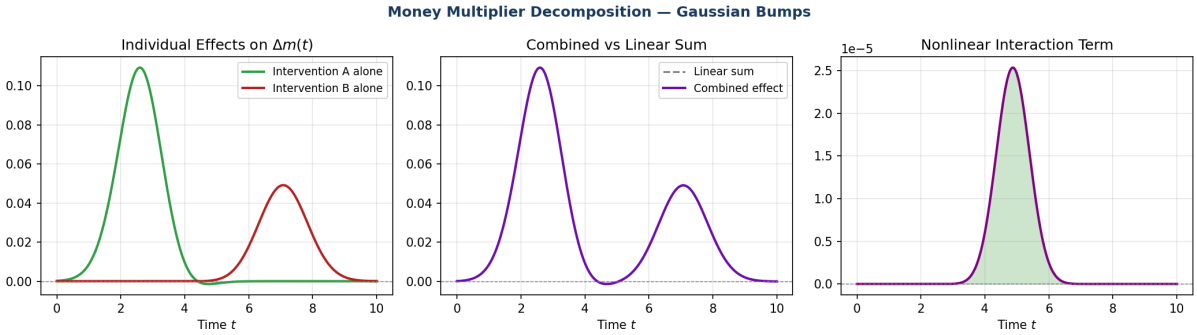

Figure 7: Multiplier decomposition for Gaussian bumps. Left: individual effects of ϕ_a and ϕ_b. Centre: combined effect versus linear sum of individual effects. Right: nonlinear interaction term $\Gamma(t)$, which is non-zero because the Gaussian tails overlap.

8 Conclusion

We have developed a monetary policy transmission framework for the RALE model using generalised bump functions. The main contributions are:

1. **Three-family bump framework** (Section 2): Gaussian, compactly-supported C^∞, and raised-cosine bumps provide a hierarchy of smoothness and localisation properties suitable for different empirical settings.

2. **Well-posedness under perturbation** (Theorem 3.1): the perturbed schedule inherits the smoothness of the bump family, and the RALE equilibrium path is uniquely determined and equally smooth.

3. **Asymmetric transmission** (Proposition 4.2): expansionary interventions have strictly larger absolute IRFs than equivalent contractionary ones, due to path-dependent equity compounding.

4. **Multiplier decomposition** (Definition 5.1, Proposition 5.2): the nonlinear interaction term $\Gamma(t)$ is exactly zero when the two compact supports do not overlap, providing an identification condition for empirical work.

5. **Long-run non-neutrality of equity** (Theorem 6.2): even temporary interventions have lasting effects on the equity-to-assets ratio through the RALE compounding mechanism, while assets and liabilities are long-run neutral.

Future work will: (i) embed the bump perturbation framework in the stochastic RALE extension, treating h_a and h_b as random variables to model policy uncertainty; (ii) calibrate the bump widths w_a, w_b to the observed duration of historical QE and QT programmes; and (iii) extend to the network RALE setting to study how bump-driven policy shocks propagate through the interbank exposure matrix.

References

[1] Ghosh, S. *The RALE Model of a Reserve Bank.* Kolkata, 2026.

[2] Ghosh, S. *The Stochastic RALE Model of a Reserve Bank.* Kolkata, 2026.

[3] Ghosh, S. *The Multi-Bank RALE Contagion Model.* Kolkata, 2026.

[4] Bernanke, B. S. and Gertler, M. (1995). Inside the black box: the credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4), 27–48.

[5] Brunnermeier, M. K. and Sannikov, Y. (2014). A macroeconomic model with a financial sector. *American Economic Review*, 104(2), 379–421.

[6] Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy.* Princeton University Press.

[7] Bernanke, B. S., Reinhart, V. R. and Sack, B. P. (2004). Monetary policy alternatives at the zero bound: an empirical assessment. *Brookings Papers on Economic Activity*, 2004(2), 1–100.

[8] Rudin, W. (1976). *Principles of Mathematical Analysis*, 3rd ed. McGraw-Hill.

[9] Hörmander, L. (1990). *The Analysis of Linear Partial Differential Operators I*. Springer.

[10] Christiano, L. J., Eichenbaum, M. and Evans, C. L. (1999). Monetary policy shocks: what have we learned and to what end? *Handbook of Macroeconomics*, Vol. 1A, 65–148. Elsevier.

Glossary of Notation

Symbol	Meaning
$R(t)$	Reserve ratio at time t
$R_{\text{base}}(t)$	Baseline (unperturbed) reserve schedule
$\widetilde{R}(t)$	Perturbed reserve schedule $= R_{\text{base}} + \phi_a + \phi_b$
$\phi_a(t)$	Bump function for intervention A, centred at $t = a$
$\phi_b(t)$	Bump function for intervention B, centred at $t = b$
c, w, h	Centre, half-width, amplitude of a bump
$a < b$	Intervention dates
$A(t)$	Total assets
$L(t)$	Total liabilities
$E(t)$	Equity (shareholders' residual claim)
E_0	Initial equity
$m(t)$	Effective money multiplier $A(t)/R(t)$
$A^*(t)$	Unique solution of the RALE fixed-point equation
$\Delta X(t)$	Impulse-response function of variable X
$\Gamma(t)$	Nonlinear interaction term in multiplier decomposition
ϕ^{G}	Gaussian bump
ϕ^{C}	Compactly-supported C^∞ bump
ϕ^{RC}	Raised-cosine bump
$G(R, t)$	Smooth function $R \mapsto A^*$ from implicit function theorem

Proof of Theorem 6.2 (Expanded)

We give a more detailed proof of part (ii) of Theorem 6.2.

For $t > a + w_a$ (after the expansionary bump has concluded for compact families), the perturbed schedule satisfies $\widetilde{R}(t) = R_{\text{base}}(t) + \phi_b(t)$. The RALE fixed-point equation under each schedule is

$$A_{\text{pert}}(t) - \widetilde{R}(t)[1 + A_{\text{pert}} - E_0(1 + A_{\text{pert}})^t] = 0,$$

$$A_{\text{base}}(t) - R_{\text{base}}(t)[1 + A_{\text{base}} - E_0(1 + A_{\text{base}})^t] = 0.$$

For $t > b + w_b$ (both bumps concluded), $\widetilde{R}(t) = R_{\text{base}}(t)$, hence $A_{\text{pert}}(t) = A_{\text{base}}(t)$ by uniqueness. Therefore $E_{\text{pert}}(T) = E_0(1 + A_{\text{pert}}(T))^T = E_0(1 + A_{\text{base}}(T))^T = E_{\text{base}}(T)$, giving $\Delta E(T) = 0$.

The *intermediate* non-neutrality (for $a < t < T$ and before $b + w_b$) arises because the path of $A_{\text{pert}}(s)$ for $s \in [a, t]$ differs from $A_{\text{base}}(s)$, and the equity $E(t) = E_0(1 + A(t))^t$ records the *current* asset level only. During the window $[a, a + w_a]$ the expansionary bump raises A_{pert} above A_{base}, which raises E_{pert} above E_{base}. After the window, reserves revert but the equity level has been durably elevated through the $(1 + A)^t$ compounding. This is the RALE memory effect. □

The End